

Subject Description Form

Subject Code	COMP 5322
Subject Title	Internet Computing and Applications
Credit Value	3
Level	5
Pre-requisite/ Exclusion	
Objectives	The objectives of this subject are to enable students: 1. to study the impact of Internet in facilitating a truly distributed, wide area and highly accessible computing environment; 2. to explore various web-related technologies and to gain appreciative knowledge of how these technologies synergize with one another to enable ubiquitous access of information; 3. to examine the analysis, design and implementation techniques required to develop the network, enterprise and Internet based information systems. In also covers the managerial perspective of Internet Computing and how this evolving technology will impact future enterprise e-solution.
Intended Learning Outcomes	After completing this subject, students should be able to: a) demonstrate a comprehensive and in-depth understanding and appreciation of advanced Internet Computing technologies and apply the skills to plan, design and formulate creative solutions for the deployment of state-of-the-art web-based information systems, addressing complex interdisciplinary issues with professionalism and, leveraging emerging technologies for global challenges; b) critically assess the complex problems and issues surrounding the challenges. In the process of formulating a holistic solution to the problems, students should apply the advanced knowledge and developed skill sets to comprehensively assess the suitability of various technologies, fostering intercultural and interdisciplinary interactions and demonstrating critical awareness and integrity in making responsible decisions; and c) show in-depth understanding of the core Internet Computing concepts that will provide them with the necessary skill sets to acquire further knowledge as the technology continues to evolve.
Subject Synopsis/ Indicative Syllabus	1. Introduction to Internet Computing for Enterprises • The role of Internet technologies in enabling distributed enterprise information systems • Evolution from traditional IS to Internet-based IS • Overview of enterprise application architectures (multi-tier, client/server, microservices)

	2. Enterprise Network Infrastructure & Support • Fundamentals of LAN/WAN, data centers, and network protocols • Internet connectivity, CDNs, and modern enterprise networking • Introduction to cloud infrastructure as enterprise backbone 3. Web Architecture & Core Protocols • Web as a universal client/server platform • HTTP/HTTPS protocols, request/response cycle • HTML, CSS, and basic web standards • Web browsers and web servers in enterprise IS 4. Web Application Development Technologies • Front-end (HTML5, CSS3, JavaScript) and back-end (CGI, servlets, modern approaches) • Dynamic content generation and asynchronous communication (AJAX, fetch API, RESTful APIs) • State management (sessions, cookies) 5. Web Security & Best Practices • Secure web application design • OWASP Top 10 threats (e.g., XSS, CSRF, SQL Injection) • HTTPS, content security policy, and secure authentication flows 6. Web Database Connectivity & Enterprise Integration • Database connectivity: web-to-database (SQL, NoSQL, middleware) • Enterprise integration via web services (REST, SOAP) and APIs • Middleware, microservices, and service-oriented architecture (SOA) 7. Managerial Perspectives and Future Directions • Digital transformation and the evolving role of IT in enterprises
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	• Change management and enterprise adoption of Internet technologies • Strategic impacts of cloud, AI, and IoT on business models • Discussion: Future directions for Internet computing and enterprise e-solutions																							
Teaching/Learning Methodology	Class activities including - lecture, tutorial, lab, workshop seminar where applicable In the lectures, the basic web building blocks and applications will be elaborated. In the tutorials, some popular CSS and Javascript frameworks such as Bootstrap and jQuery are be introduced. The students are required to develop their personal website as the midterm projects and develop an internet application as the final project.																							
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>Assignments, Tests & Projects</td><td>30</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100</td><td></td><td></td><td></td></tr></table>	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	Assignments, Tests & Projects	30	✓	✓	✓	Final Examination	70	✓	✓	✓	Total	100			
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Student study effort expected	<table><tr><td colspan="2">Class Contact:</td></tr><tr><td>Class activities (lecture, tutorial, lab)</td><td>39 hours</td></tr><tr><td colspan="2">Other student study effort:</td></tr><tr><td>Assignments, Tests, Projects, Exams</td><td>66 hours</td></tr><tr><td>Total student study effort</td><td>105 hours</td></tr></table>	Class Contact:		Class activities (lecture, tutorial, lab)	39 hours	Other student study effort:		Assignments, Tests, Projects, Exams	66 hours	Total student study effort	105 hours													
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Reading list and references	[1]. James F. Kurose and Keith W. Ross, <i>Computer Networking: A Top-Down Approach</i>, 8th ed. Hoboken, NJ: Pearson, 2021. [2]. W3Schools.com, “W3Schools Online Web Tutorials,” 2025. [3]. Andrew Hoffman, <i>Web Application Security: Exploitation and Countermeasures for Java, Ruby, and Python</i>, 2024.																							